

Developing New Packages

In case no package already exists for one's requirement, one can develop new custom packages. In this case, one must minimize the risk of version collision by specifying the version range in the pubspec file.

The `url_launcher` plugin package enables one to open the default browser on the mobile platform for displaying a particular URL. It is supported on both iOS and Android. This package shows that even packages can contain platform-specific codes. Such packages are called plugins.

There are basically two types of packages:

- **Dart packages:** Dart packages include basic packages which are written in Dart. An example is the `path` package. Some of these packages contain Flutter specific functionality and are hence dependant on the Flutter framework. It is this very reason why they can only be used on the Flutter platform. A good example is a `fluro` package whose usage is restricted only to the Flutter platform.
- **Plugin packages:** A plugin package consists of a specialised Dart package which contains an API written in a Dart code. It also contains a platform-specific implementation for Android and iOS. A good example of a plugin package is the `battery` plugin package.

Developing a Dart package is comparatively simpler than developing a plugin one. If one wants to develop a package which calls into platform-specific API, a plugin package is what one is looking for. It is a specialised version of the Dart package. Here, the API is connected to the platform-specific implementation using platform channels.

What Runs Behind It: Background Processes

Flutter enables one to execute Dart code in the background. The feature is made possible by the setting up of an isolate. Now, these isolates are Dart's model for multithreading. An isolate, however, differs from a conventional thread because it does not share a memory with the main program. One has to set up one's isolate for executing the Dart code in the background using callback and callback dispatcher.

Flutter Package Site: All the Packages

The Flutter Package site displays a number of packages which are compatible with Flutter. Some of the most common packages that are used extensively are listed here.